

# Introduction to Secure Multi-Party Computation (SMPC)

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# Agenda

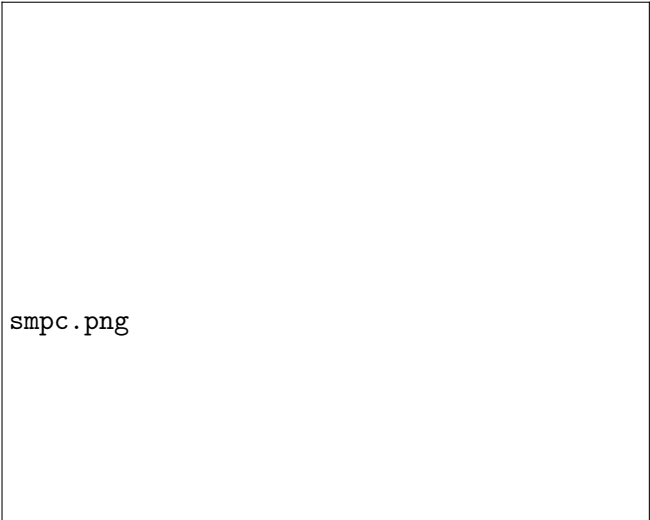
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## Motivation

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Several parties aim to compute a shared outcome without disclosing their private inputs.

Example: Obtain the average salary within a team without disclosing the individual salaries

A large rectangular box with a thin black border, representing a diagram or image placeholder.

`smpc.png`

## Requirements

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### 1. Privacy

No party can learn any information about the others besides the output

### 2. Correctness

Each party is guaranteed to get the correct output at the end of the computation

### 3. Independence of inputs

Each party can independently choose their input

### 4. Guaranteed output delivery

Malicious parties cannot block honest parties from obtaining the output

### 5. Fairness

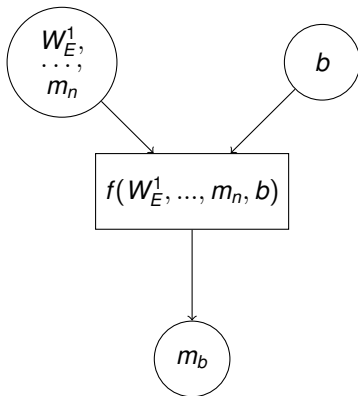
All parties get access to the output

## Oblivious Transfer

Fundamental building block for many SMPC protocols

- Transfer one message from a set of  $n$  messages
- Sender does not know which message was chosen
- Receiver learns only the chosen message

Implementation using asymmetric cryptography



## Additive secret sharing [geeksforgeeks\_smpc]

### Roles

$n$  participants

### Threat model

semi-honest

less than  $n - 1$  parties  
collude

Extension to malicious  
adversaries: MAC

1. Each party splits their data  $d_i$  into  $n$  parts  $s_1, \dots, s_n$ , such that  $\sum_{j=1}^n s_j = d_i$  and sends  $s_i$  to party  $i$
2. Each party computes the sum of the received values and shares it with the other parties
3. Each party computes the sum of the received sums, which is equivalent to the sum of all  $d_i$

## Yao's garbled circuits: Secure 2-party computation [giappppp\_isogeny\_nodate]

### Roles

Garbler G and  
Evaluator E

Threat Model  
semi-honest

1. G picks random labels  $W_G^0, W_G^1, W_E^0, W_E^1$  corresponding to  $g = 0, g = 1, e = 0, e = 1$ .
2. G creates the permuted garbled output and shares with E
3. G knows  $g$  and therefore transmits  $W_G^g$
4. Access to more than one input-output-relation would reveal information about  $g$ . To ensure that G doesn't learn  $e$  and E only learns one ciphertext,  $W_E^e$  is transmitted using Oblivious Transfer
5. E can now decrypt the output and shares it with G

| $g$ | $e$ | output $g \wedge e$ | garbled output            | permuted garbled output   |
|-----|-----|---------------------|---------------------------|---------------------------|
| 0   | 0   | 0                   | $Enc(H(W_G^0, W_E^0), 0)$ | $Enc(H(W_G^0, W_E^1), 0)$ |
| 0   | 1   | 0                   | $Enc(H(W_G^0, W_E^1), 0)$ | $Enc(H(W_G^1, W_E^1), 1)$ |
| 1   | 0   | 0                   | $Enc(H(W_G^1, W_E^0), 0)$ | $Enc(H(W_G^0, W_E^0), 0)$ |
| 1   | 1   | 1                   | $Enc(H(W_G^1, W_E^1), 1)$ | $Enc(H(W_G^1, W_E^0), 0)$ |

## Limitations

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- Computational overhead
  - Limited applicability on IoT or mobile devices
- High communication costs / latency
  - Infeasible for real-time applications
  - Scalability issues
- Vulnerable to dishonest majority scenarios
- Strict data structure requirements



## 1-2 Oblivious Transfer [chou2015simplest]

- Ginny holds two messages  $W_E^0, W_E^1$  and wants Evan to learn exactly one of them.
- She generates an RSA key pair  $(N, e, d)$  and chooses two random values  $x_0, x_1$ . She sends  $(N, e, x_0, x_1)$  to Evan.
- Evan selects his input  $e \in \{0, 1\}$  and picks  $x_e$ . He samples a random value  $k$  and computes

$$v = (x_b + k^e) \bmod N.$$

He sends  $v$  to Ginny.

- Ginny computes

$$k_0 = (v - x_0)^d \bmod N, \quad k_1 = (v - x_1)^d \bmod N.$$

Exactly one of these equals Evan's  $k$ ; the other is pseudorandom from Evan's perspective.

## 1-2 Oblivious Transfer [chou2015simplest]

- Ginny masks her messages using the two keys:

$$W_E^{0'} = (W_E^0 + k_0) \bmod N, \quad W_E^{1'} = (W_E^1 + k_1) \bmod N,$$

and sends  $(W_E^{0'}, W_E^{1'})$  to Evan.

- Evan recovers his chosen message by computing

$$m_e = (m_e' - k) \bmod N.$$

Without knowing  $d$ , he cannot derive the other key and thus cannot obtain  $W_E^{1-e}$

## References

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